Polynomials

Questions 1–28

Question 1. Expand each of the following:

a) $(x-1)(2x+1)(7x+3)$
b) $(2x-1)(3x-4)(5x-2)$
c) $(4x+3)(2x+1)(3x-2)$

Question 2. Fully factorise each of the following:

a) $8x^3 - 12x^2 - 28x$
b) $10x^4 - 20x^3 - 30x^2$
c) $6x^4 - 45x^3 - 24x^2$

Question 3. Fully factorise each of the following:

a) $2x^2y^3z - 6xy^4z^2$

b) $4x^2y^3z^4 + 14x^3y^2z^5$

c) $3(x-1)^3y^2z - 9(x-1)^2yz^3$

d) $9x^2(2y-1)^3z - 15x^3(2y-1)z^2$

e) $8x^2(3y-2)^3(z+1) - 4x^3(3y-2)(z+1)^2$

Question 4. State the degree of the following polynomials:

a) $6x^4 - 9x^2 + 2$

b) $4x^2 - 3x^5 + 1$

c) $8 + 4x^3 - x^6$

Question 5. Expand each of the following:

a) $(x+2)(x^2-3x+1)$
b) $(2x-1)(x^3+2x^2-x+3)$
c) $(x^2+2x+3)(3x^2-x+2)$
d) $(2x^2-4x+1)(x^2+3x-2)$

Question 6. Simplify the fractions:

a) $\frac{3x^3+5x^2}{x}$
b) $\frac{7x^3+9x^{10}}{3x}$
c) $\frac{(x+1)(x-5)}{x+1}$
d) $\frac{(x-2)^3}{x-2}$
e) $\frac{x+4}{(3x-1)(x+4)}$
f) $\frac{x-5}{2x^2-9x-5}$
g) $\frac{2x^2-15x+7}{2x^2+5x-3}$

Question 7. Divide

a) $4x^3 - 22x^2 + 34x - 12$ by $2x - 4$
b) $x^4 + 6x^3 + 7x^2 + x + 6$ by $x + 2$
c) $6x^5 - 15x^4 + 9x^3 - 36x^2 + 72x - 24$ by $3x - 6$
d) $32x^4 - 12x^3 + x^2 + 20x - 5$ by $4x - 1$
e) $2x^6 + x^5 - 3x^4 + 4x + 6$ by $2x + 3$
f) $x^5 + 3x^4 + 4x^3 - 2x^2 - 4x - 5$ by $x^2 + x + 1$
g) $x^7 - 5x^6 + x^5 - 4x^4 - 8x^3 + 13x^2 - 8x + 12$ by $x^2 + 1$

Question 8. Divide

a) $x^3 + 9x^2 + 19x + 6$ by $x + 1$
b) $4x^3 - 10x^2 - 12$ by $2x - 1$
c) $x^4 + 2x^3 - x^2 + x + 16$ by $x + 2$
d) $x^5 - 4x^3 + x^2 + 10$ by $x^2 - 1$

Question 9. Given $(x-5)$ is a factor of $f(x)$, where $f(x) = x^3 - 3x^2 - 8x - 10$,

a) Write $f(x)$ as a product of a quadratic and a linear expression.

b) Show that $f(x) = 0$ only has one real solution.

Question 10. Given $(x+2)$ is a factor of $f(x)$, where $f(x) = 2x^3 - 7x^2 - 10x + 24$,

a) Write $f(x)$ as a product of a quadratic and a linear expression.

b) Fully factorise $f(x)$.

c) Find the solutions to $f(x) = 0$.

d) Write down the roots of $f(x-2) = 0$.

Question 11. Given $(2x + 1)$ is a factor of $f(x)$, where $f(x) = 2x^3 - x^2 + 35x + 18$,

a) Write $f(x)$ as a product of a quadratic and a linear root.
b) Show that $f(x) = 0$ only has one real solution.
c) State the one real solution to $f(x) = 0$.
d) Write down the real root of $f(x - 4) = 0$.

Question 12. Given $f(x)$ and $g(x)$ both have a common factor of $(x - 1)$ where $f(x) = x^3 + 4x^2 - 5x$ and $g(x) = x^3 - 3x + 2$,

a) Fully factorise $f(x)$ and $g(x)$.
b) Show that $(x - 1)$ is also a factor of $f(x) - g(x)$.

Question 13. The Factor Theorem Use the factor theorem to prove that

a) $(x-3)$ is a factor of $x^3 - 10x^2 + 13x + 24$

b) $(x+5)$ is a factor of $x^3 - 5x^2 - 10x + 200$

c) $(4x-1)$ is a factor of $4x^3 - 21x^2 + 29x - 6$

Question 14. In each case, use the factor theorem to prove the given factor is a factor and hence fully factorise the expression:

a) $(x-6)$ is a factor of $2x^3 - 11x^2 - 9x + 18$

b) $(2x+5)$ is a factor of $6x^3 + 19x^2 + 8x - 5$

Question 15. It is given that $(x+1)$ is a factor of $(x+3)(x^2-kx+4)-20$ where k is a positive constant. Find the value of k.

Question 16. It is given that $(x+3)$ is a factor of $f(x)=2x^3+7x^2-px-18$ where p is a positive constant.

a) Find the value of p.

b) Fully factorise $f(x)$.

Question 17. It is given that $(8x + 15)$ is a factor of $f(x) = 32x^3 + px^2 + 219x + 45$ where p is a positive constant.

a) Find the value of p.

b) Fully factorise $f(x)$.

Question 18. It is given that $(x + 1)$ is a factor of $f(x) = 2x^3 + 11x^2 - px - 2p$ where p is a positive constant.

a) Find the value of p.

b) Fully factorise $f(x)$.

Question 19. Given that $(x+2)$ and $(x-5)$ are factors of $f(x) = px^3 - 5x^2 - 23x - q$ where p and q are positive constants.

a) Find the values of p and q.

b) Fully factorise $f(x)$.

Question 20. Given that $(x+3)$, $(x-1)$ and $(2x-1)$ are factors of $f(x) = px^4 + qx^3 + rx^2 - 37x + 15$ where p, q and r are constants. Find the values of p, q and r.

Question 21. It is known that $(x+9)$ is a factor of $f(x) = 3x^4 + 59x^3 + px^2 + 369x - 162$.

a) Find the value of p.

b) Fully factorise $f(x)$.

Question 22. It is given that $x^2 + bx + c$ has a factor of $(x-1)$ and $x^2 + px + q$ has a factor of $(x+2)$. Show that $5 + b + c - 2p + q = 0$.

Question 23. Show that $135k^3 + 270k^2 + 165k + 30$ is always a multiple of 30 when k is a positive, whole number.

Question 24. Prerequisite knowledge for Q24-Q28: Graph Sketching $f(x) = x^3 + 2x^2 + 2x - 20$

a) Use the factor theorem to show that $(x - 2)$ is a factor of $f(x)$.

b) Prove that the graph of $f(x)$ intersects the x-axis only once and state its coordinates.

c) Given that the point of inflection is to the left of the root of the curve, sketch $y = f(x)$.

Question 25. It is given that $(x+3)$ is a factor of $f(x) = x^3 - 7x^2 - px + 75$ where p is a positive constant.

a) Find the value of p.

b) Find the solutions to $f(x) = 0$.

c) Sketch the graph of $y = f(x)$.

Question 26. Given that $(x-3)$ and $(x+1)$ are factors of $f(x) = x^3 - (2+k)x^2 + (2k-3)x + 3k$ where $0 < k < 3$. Sketch the graph of $f(x)$.

Question 27. It is given that $f(x) = x^3 - (2k+4)x^2 + (k^2+8k)x - 4k^2$ where k is a positive constant.

a) Prove that $(x-4)$ is a factor of $f(x)$.

b) Given $0 < k < 4$, sketch the graph of $f(x)$.

Question 28.

a) Divide $x^3 - x^2 + 5$ by $x - 1$.

b) Hence, sketch the graph of $y = \frac{x^3 - x^2 + 5}{x-1}$.